Quantifying STEM-mode spectra using ζ-factors

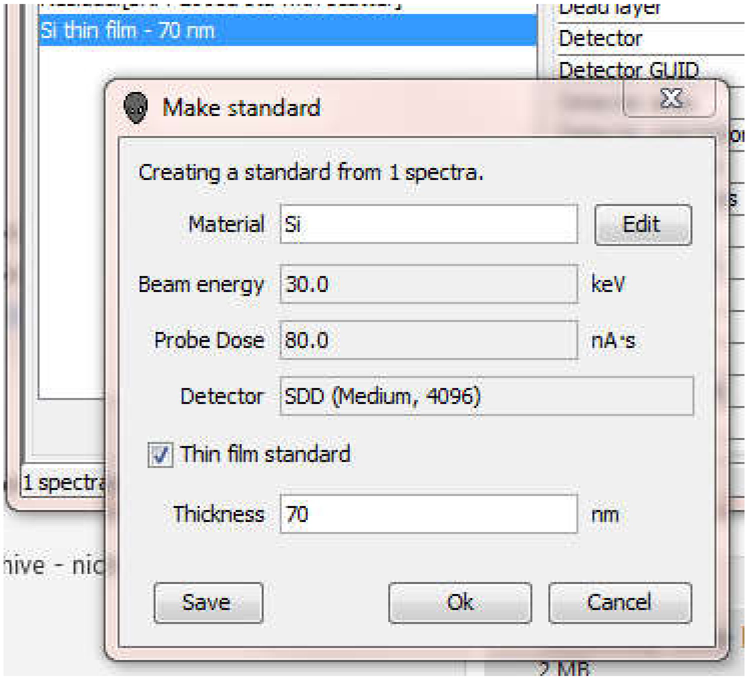
Getting ready

You will need the following information to perform a STEM-mode quantification. STEM, of course, means scanning transmission electron microscopy and the samples were are talking about are unsupported thin-films or particles on an essentially transparent thin film – no substrate.

1. Standard spectra collected from thin films of known composition and mass-thickness for each element in the unknown.
    a. Maybe the single largest challenge in STEM quant is finding suitable artifacts to act as standards. Usually, standard thin-films are prepared and then characterized by other techniques.
    b. Standards maybe single element or multi-element but there may not be any interferences between elements in the standard.
    c. The beam dose (live time · probe current) must be known for each standard spectrum
    d. The standard thin-films should be thin enough that less than a percent or two of the incident beam energy is lost in the film.
    e. The standards should have been collected at the same beam energy as the unknown spectrum,
2. Reference spectra collected from bulk or thin samples for any spurious peaks like Cu from the grid.
    a. The mass-thickness or the probe dose is not necessary for reference spectra.
3. The unknown spectrum along with the probe dose.
    a. The unknown film should be thin enough that less a percent or two of the incident beam energy is lost in the film.
    b. If the unknown and standard spectra are collected without interruption on a stable instrument, it may not be necessary to measure the probe current so long as it remains constant.

Prepare the standard spectra

1. It is easiest if standard spectra contain all the information required to facilitate the quantification. DTSA-II provides a mechanism to ensure that this data is encapsulated within the DTSA-II-extended EMSA spectrum files. Your spectra files should contain data specifying the:
    a. Probe current
    b. Live time
    c. Composition
    d. Mass-thickness
    e. Beam energy
    f. Detector
2. Fortunately DTSA-II provides the "make 'Standard Bundle'" dialog to ensure that spectra contain all the necessary data.



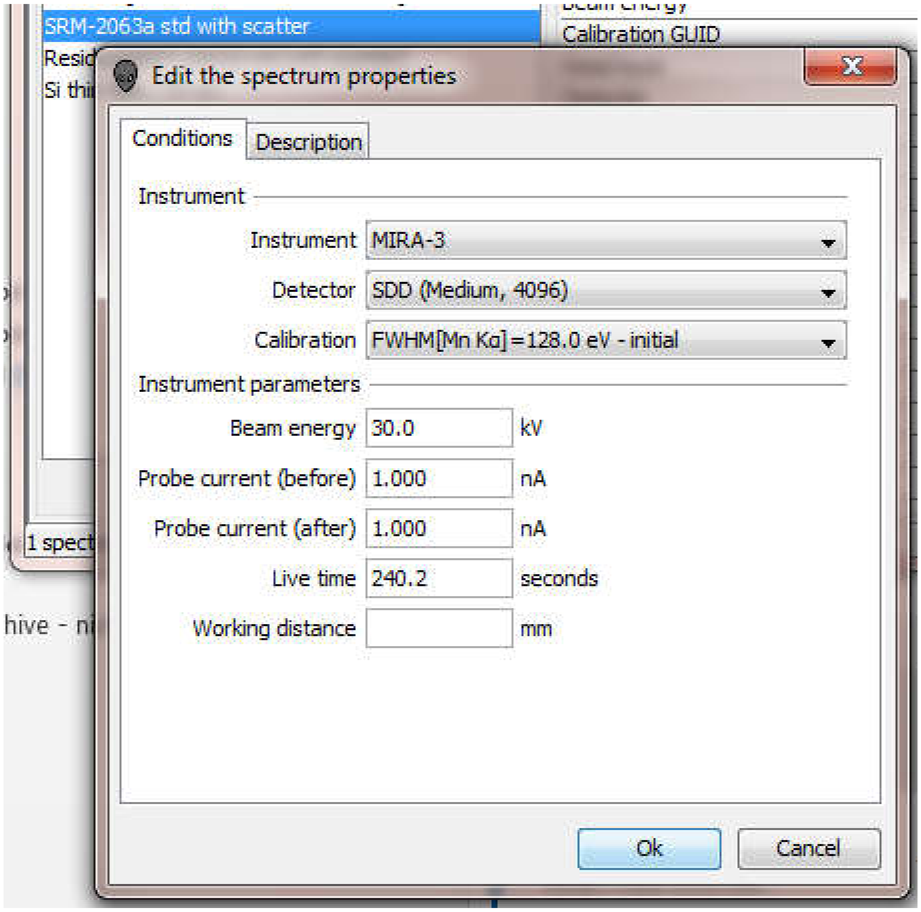
The "make standard" dialog is accessed by selecting a spectrum (or multiple spectra if you wish to combine them) in the "spectrum list" and selecting the "make standard" option from the "spectrum list" right-click popup menu. Since we wish to perform thin film quant, we will need to specify a material with a density and the thickness of the film.

3. Use the "save" button in the lower, left of the "make standard" dialog to write the spectrum to disk. Remember where you saved it as you will need to reload this spectrum from within the "quantification wizard."
4. Both the unknown and standards should be associated with the same instrument and detector.

Prepare the unknown spectra

1. It is also helpful to prepare the unknown spectra. The data required for the unknown include:
    a. Probe current
    b. Live time
    c. Beam energy
    d. Detector
2. You can enter this information using the "edit spectrum properties" dialog. This is accessed from the "spectrum list" right-click popup menu or from the "Tools" item on the main menu.

Again you should select the spectrum you wish to edit in the “spectrum list.”



3. Ensure that the data on the “conditions” tab is correct and select “Ok.” You can write the spectrum to disk as an EMSA file but you should also retain the unknown spectrum in the “spectrum list.”

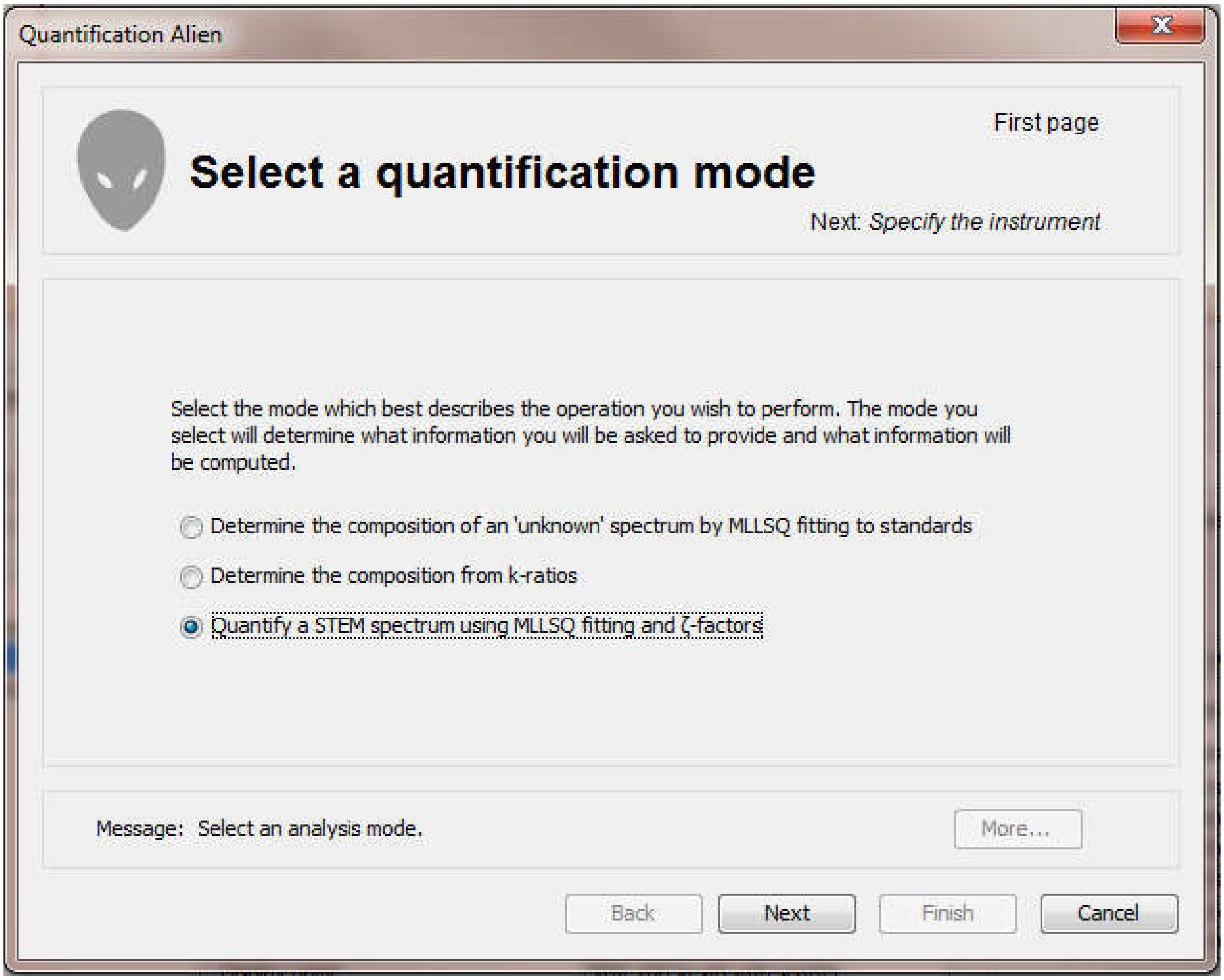
Preparing reference spectra

1. Reference spectra are used to strip spurious elements which are common in STEM spectra. Since the spectra are only used for the peak shape, bulk or thin film spectra will work and very little information is actually required. The only additional piece of information that it is useful to include in a reference spectrum is the composition of the material from which it was collected.
2. The “assign material” menu item in the “spectrum list” popup menu can be used to assign a material to the reference. This helps the program know which elements the spectrum can be used as a reference for.

Quantifying the unknown

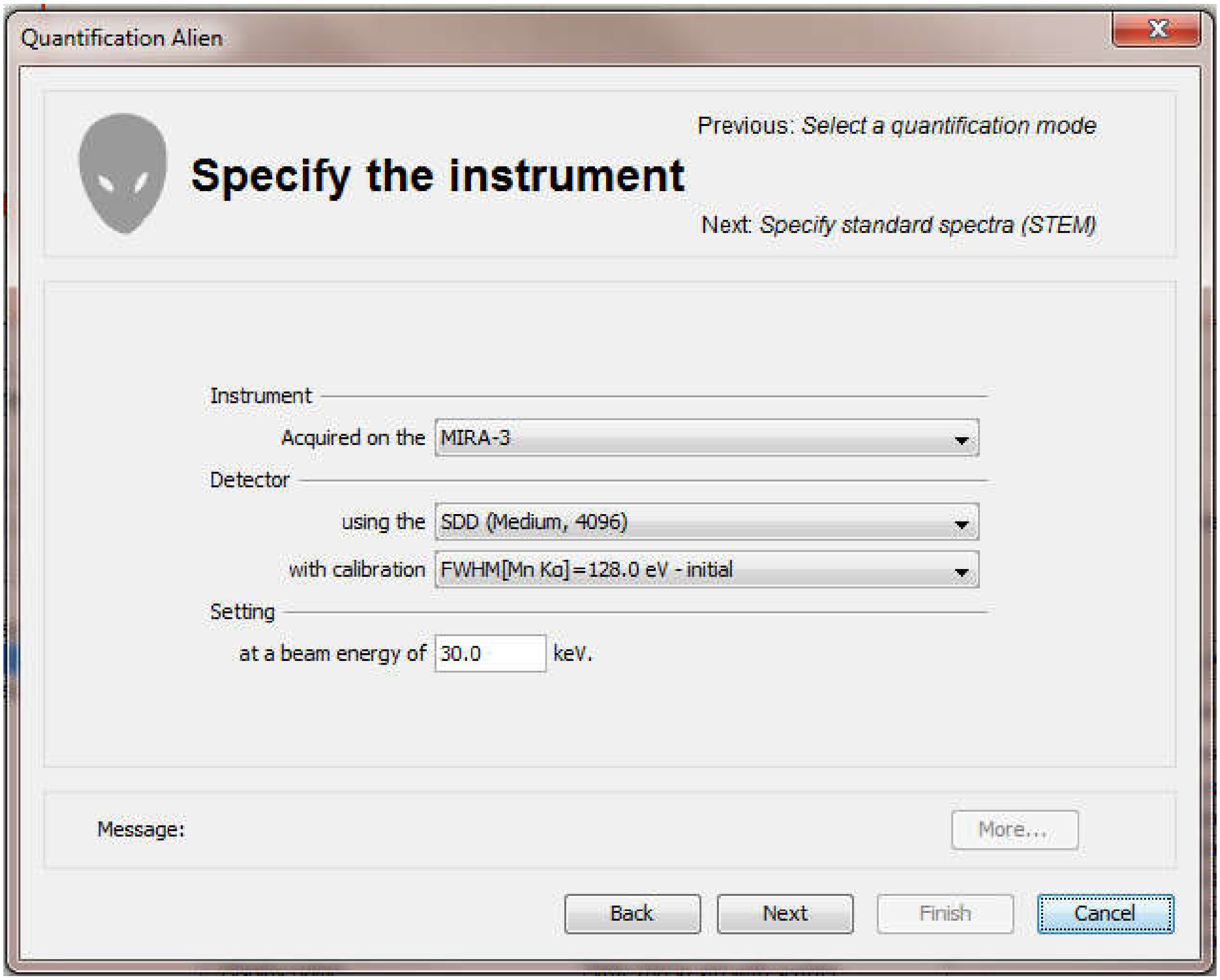
1. Select one or more edited thin-film spectra to quantify in the “spectrum list”.

2. Select the “quantification alien…” menu item from the “tools” main menu.



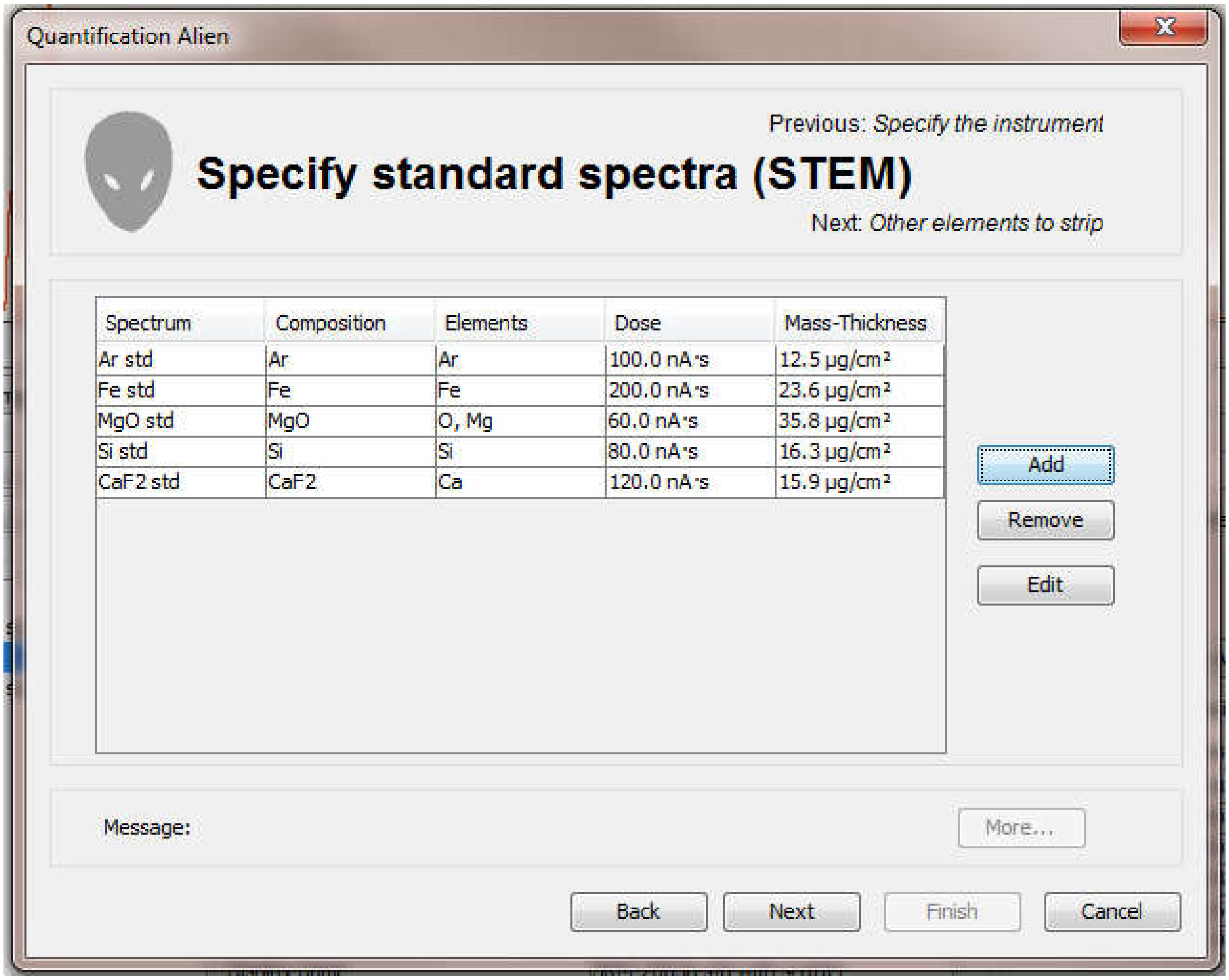
3. Select the “quantify a STEM spectrum using MLLSQ fitting and ζ-factors” option.

4. Ensure the instrument, detector, calibration and beam energy are correct. They should be as this information is read from the s

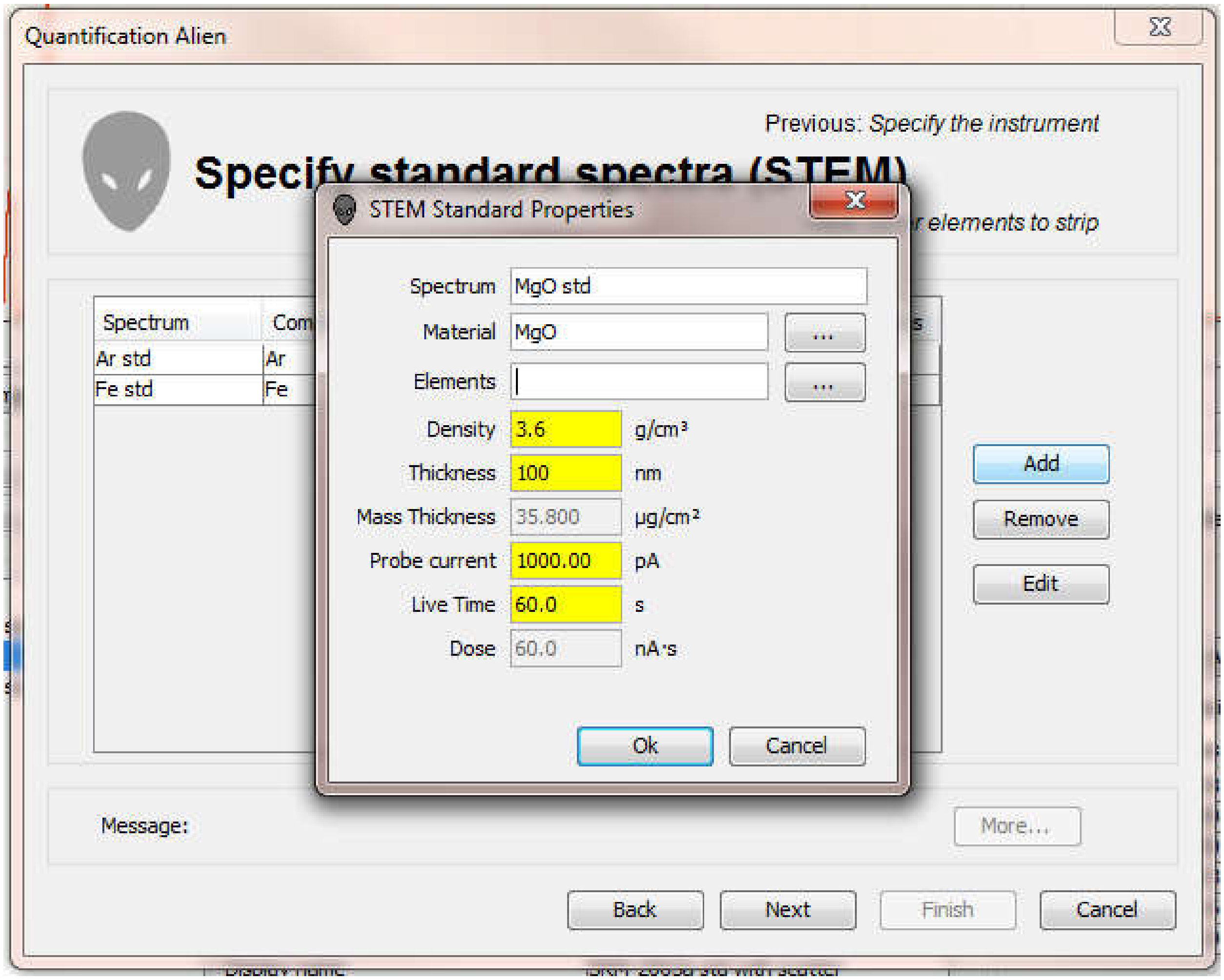


5. The next panel allows you to assign standard spectra to elements. Use the “add” and “remove” buttons to open standard spectra from disk. The “edit” button allows you to edit the properties

of the selected standard spectrum.

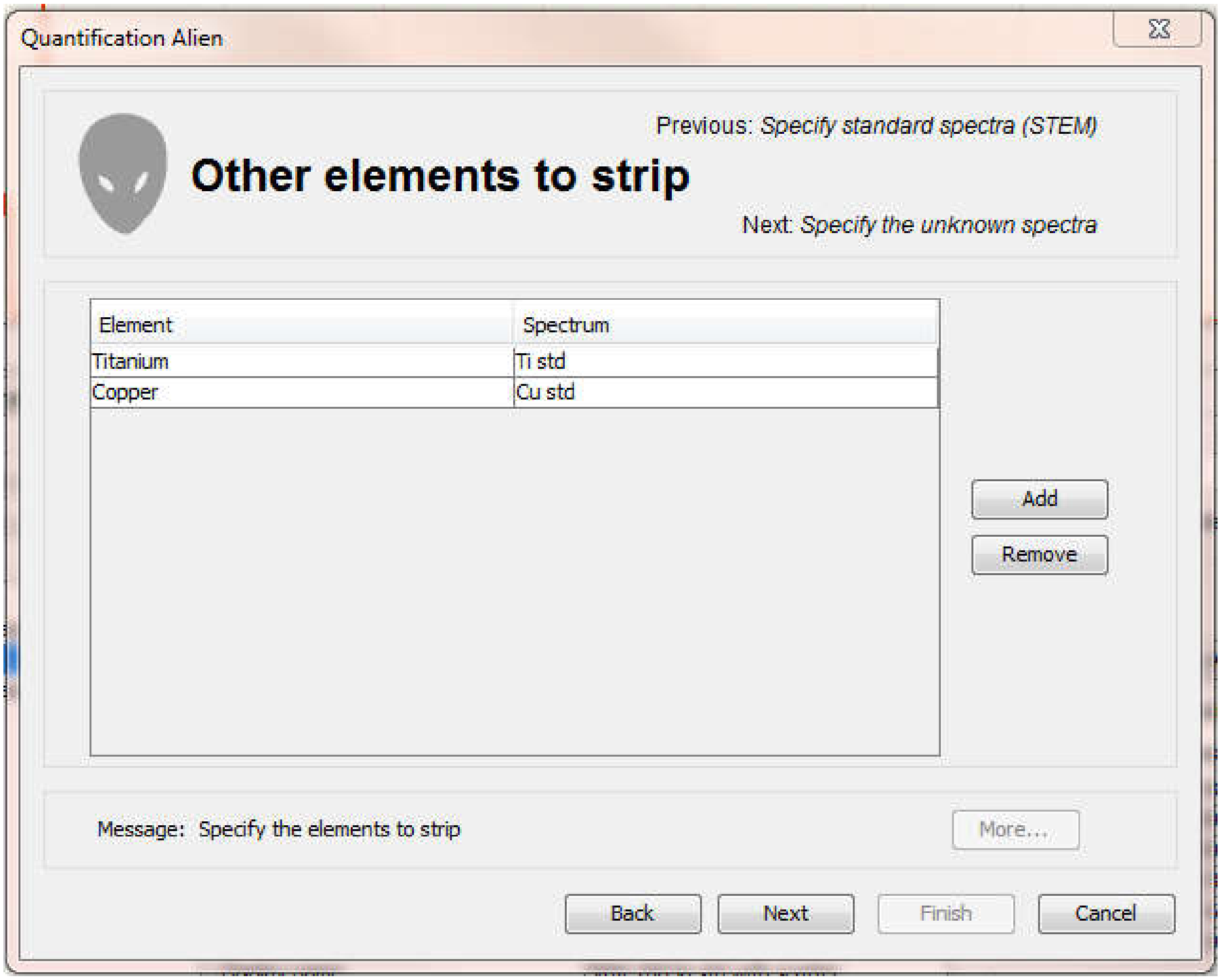


6. You may be asked to specify which element to associate multi-element spectra with. In this case, MgO could be a standard for either Mg or O or both Mg and O. Type in the elements separated by commas or spaces in the "elements" edit box or use the "…" button to select the elements from a periodic table.



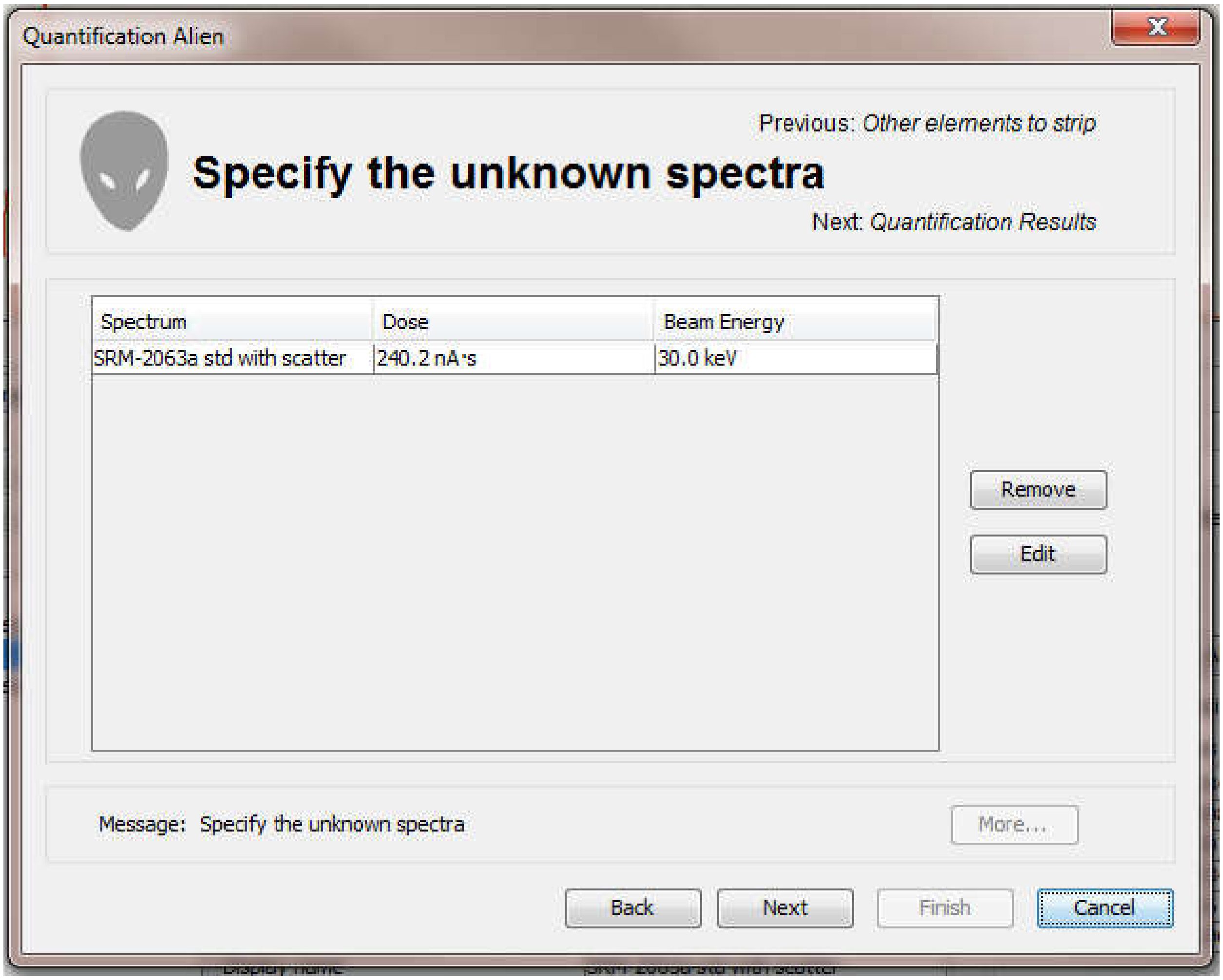
Typing “Mg, O” assigns the “MgO std” to both elements.

7. The next panel allows you to assign spectra to strip zero or more elements from the spectrum. Stripping elements is particularly important if a spurious characteristic peak from a mounting grid or a sample support is present in the spectrum and this peak interferes with an element you wish to measure.

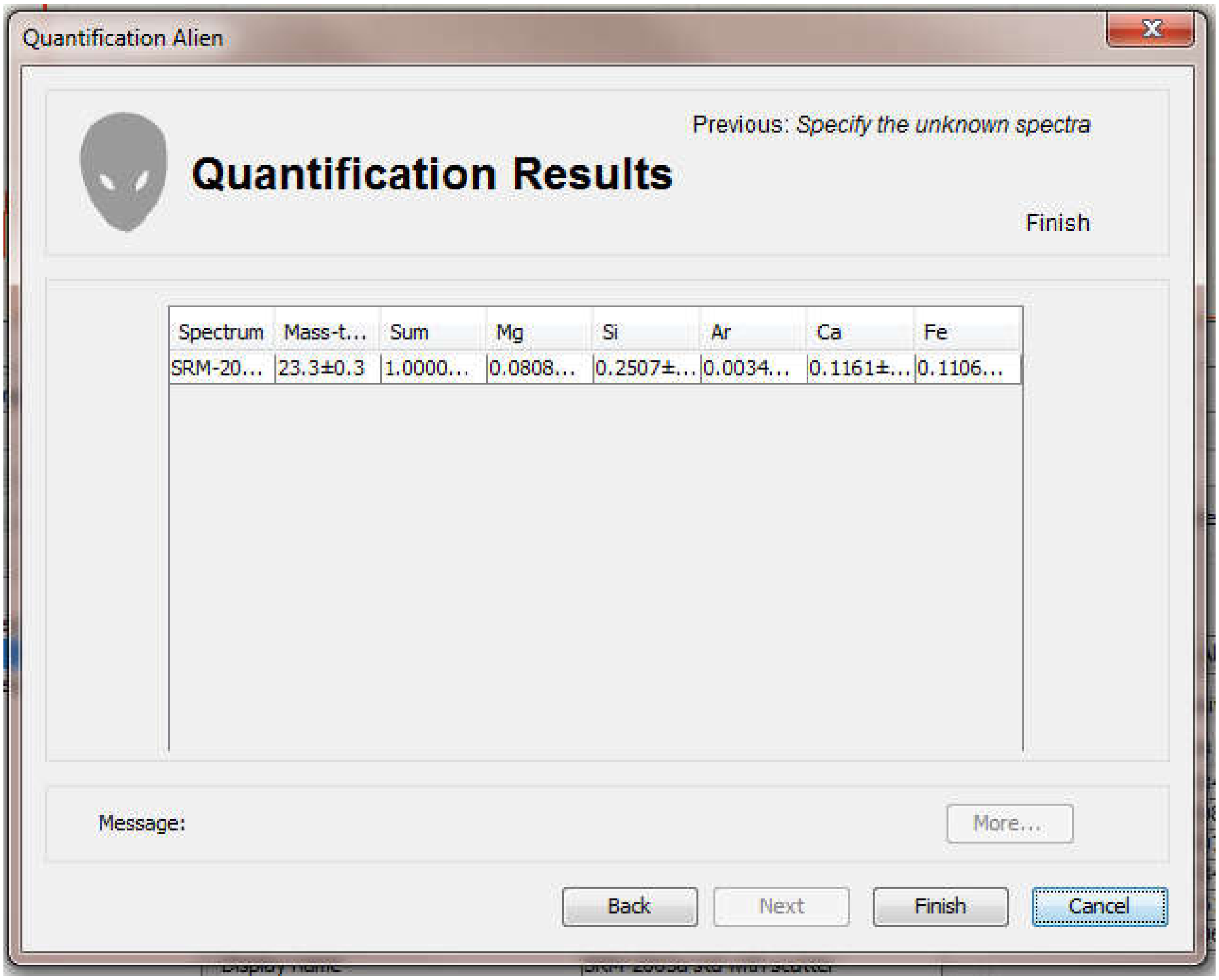


My STEM has Ti in the holder and Cu in the grid which are visible in the measured spectrum.

8. The next panel allows you to review the spectra being quantified and edit the spectrum properties.

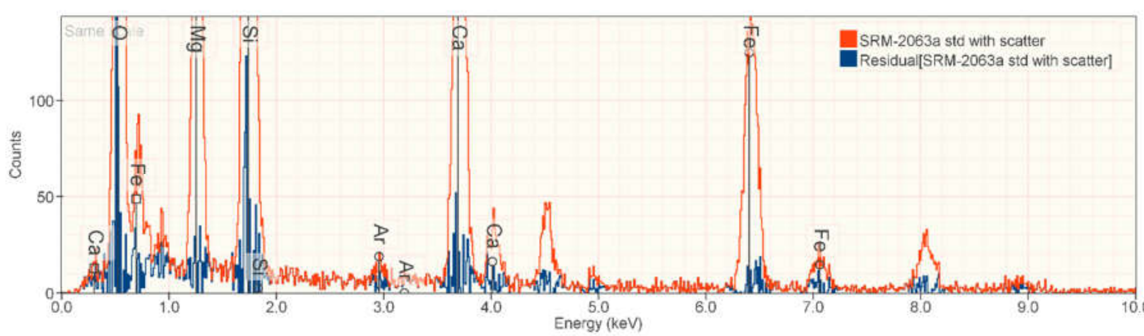


9. The final panel performs the quantification and summarizes the results.



Select the "finish" button to exit the "quantification alien."

10. In addition to the original spectra, you will find a second spectrum corresponding to the residual spectrum in the "spectrum list." Examining the residual spectrum is a good way to determine if the quantification account for all the elements in the unknown.



This residual shows that we've successfully accounted for all the elements in the unknown including the unlabeled Ti and Cu instrument scatter peaks.

11. The “report” tab contains detailed description of the quantification and the results.

Conditions

Item	Value
Instrument	MIRA-3
Detector	SDD (Medium, 4096)
Beam Energy	30.0 keV
Correction Algorithm	ζ-factors
Mass Absorption Coefficient	NIST-Chantler 2005
Stripped elements	Ti, Cu

The conditions table summarizes many instrumental parameters.

Standards

Element	Material	Element	Mass Fraction	Mass Fraction (normalized)	Atomic Fraction	Dose	Mass-Thickness $\mu g/cm^2$	Spectrum
O	MgO	O	0.3970	0.3970	0.5000	60.000	35.800	MgO std
O	MgO	Mg	0.6030	0.6030	0.5000			
Mg	MgO	O	0.3970	0.3970	0.5000	60.000	35.800	MgO std
Mg	MgO	Mg	0.6030	0.6030	0.5000			
Si	Si	Si	1.0000	1.0000	1.0000	80.000	16.310	Si std
Ar	Ar	Ar	1.0000	1.0000	1.0000	100.000	12.500	Ar std
Ca	CaF2	F	0.4867	0.4867	0.6667	120.000	15.900	CaF2 std
Ca	CaF2	Ca	0.5133	0.5133	0.3333			
Fe	Fe	Fe	1.0000	1.0000	1.0000	200.000	23.610	Fe std

The standard panel summarizes the standard spectra and the materials from which they were collected.

Results

Spectrum	Quantity	O			Mg			Si			Ar			Ca			Fe			Sum
SRM-2063a std with scatter	Line	O All			Mg All			Si All			Ar K-family			Ca K-family			Fe Kα			
	k-ratios	0.7081	±	0.0059	0.0871	±	0.0013	0.3530	±	0.0030	0.0063	±	0.0010	0.3309	±	0.0059	0.1093	±	0.0032	
Mass-thickness 23.3±0.3 μg/cm²	mass fraction	0.4384	±	0.0017(K) 0.0008(MAC) [0.0018]	0.0808	±	0.0001(K) $7.3\cdot10^{-5}$(MAC) [0.0002]	0.2507	±	0.0012(K) 0.0003(MAC) [0.0012]	0.0034	±	0.0005(K) $5.0\cdot10^{-6}$(MAC) [0.0005]	0.1161	±	0.0005(K) 0.0002(MAC) [0.0005]	0.1106	±	0.0017(K) 0.0002(MAC) [0.0017]	1.0000
I = 1.000 nA	norm(mass fraction)	0.4384	±	0.0018	0.0808	±	0.0002	0.2507	±	0.0012	0.0034	±	0.0005	0.1161	±	0.0005	0.1106	±	0.0017	-
LT = 240.2 s	atomic fraction	0.6142	±	0.0026	0.0746	±	0.0001	0.2000	±	0.0010	0.0019	±	0.0003	0.0649	±	0.0003	0.0444	±	0.0007	
Residual	C:\Users\nritchie.NIST\AppData\Local\NIST\NIST DTSA-II Reports\2015\November\6-Nov-2015\residual1436558380414768936.msa																			
Notes	Uncertainties are 1 σ and labeled by source. (K: k-ratio, MAC: mass absorption coefficient, []: combined)																			

The results table provides a detailed results report including the measured k-ratios, the measured mass-thickness, the measured mass-faction which is also expressed in normalized and atom fraction forms. The blue links allow the original spectrum and residual to be reloaded from disk.

The report provides uncertainties which are based on count statistics and assumed uncertainties in the mass absorption coefficients.

Nicholas W. M. Ritchie
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